

Admission Gates as a Quality-Control Layer for LLM-Driven Alpha Mining

An Injected-Truth Validation on 795 CSI 800 Constituents

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Abstract

LLM-driven frameworks such as AlphaSeek and QuantaAlpha excel at mining alpha factors, but none reports the probability that a discovered factor is a false discovery. We present a set of **admission gates**—statistical criteria that must be satisfied before a mined conclusion is adopted—that operates as a *quality-control layer on top of* any alpha mining framework. We validate the gates through an **injected-truth protocol** with a **random-admission control** on 795 CSI 800 constituents (2019–2026). Applied to a 72-configuration grid search (simulating the output of an alpha mining framework), our gates reduce the adoption rate from 100% to 5%, eliminate all false discoveries in the weak-signal regime, and correctly identify that the training-best configuration (Calmar 1.04) fails out-of-sample (Calmar 0.65). We further discover a **temporal aggregation mismatch**: per-trade signal improvements of +7.7pp do not transfer to portfolio-level performance (+7.0% vs. buy-and-hold +20.3%) because signals cluster near rebounds while portfolios hold through troughs. A 30-trial Bayesian optimisation (TPE) confirms the finding with 58% fewer trials. We position admission gates as a complementary layer: AlphaSeek mines, we validate.

1 Introduction

LLM-driven frameworks are transforming alpha factor discovery. AlphaSeek [AlphaSeek, 2026] mutates mining trajectories and reports CSI300 ARR of 8.28%; QuantaAlpha [Han et al., 2026] evolves trajectories via self-reflection with ARR 27.75%; FactorMiner [Wang et al., 2026] achieves IC 8.25% on CSI500; AlphaAgent [AlphaAgent, 2025] constrains generation by syntactic originality. These systems are powerful. But they share a blind spot: *none reports the probability that a discovered factor is a false discovery.*

The replication crisis in finance is well documented: McLean and Pontiff [2016] showed returns to published predictors decline 58% post-publication; Harvey et al. [2016] showed the appropriate t -threshold rises with the number of factors tried; Hou et al. [2020] found 82% of published anomalies fail multiple-testing hurdles. These results imply that faster mining without better validation produces *more false discoveries*, not fewer.

We propose admission gates as a **quality-control layer** that operates on top of any alpha mining framework. The gates do not mine—they *validate*. Given a set of candidate signals produced by AlphaSeek, QuantaAlpha, or any other framework, our gates determine which conclusions are statistically trustworthy enough to deploy.

1.1 Contributions

1. **Injected-truth protocol with random-admission control:** We validate admission gates by injecting known- α signals, using a control that adopts at the same rate. Gates eliminate false discoveries in the weak-signal regime (0% false positive rate).
2. **Temporal aggregation mismatch:** We discover that per-signal improvements of +7.7pp do not transfer to portfolio-level returns (+7.0% vs. +20.3% buy-and-hold), because signals cluster near rebounds while portfolios hold through troughs.
3. **BO underperforms grid for threshold signals:** TPE with 30 trials achieves 93.6% of the grid optimum with less overfitting, but cannot find the grid optimum for discontinuous response surfaces.
4. **M-series rules:** We distil 12 rules from master traders (Simons, Thorp, de Prado, top Chinese funds) into executable checks, each mapped to a concrete failure case from live operation.

2 Related Work

Replication crisis. ? argued most research findings are false. McLean and Pontiff [2016] showed 58% post-publication decline. Harvey et al. [2016] showed $t > 3.0$ needed. Hou et al. [2020] found 82% of anomalies fail.

Statistical machinery. White [2000] Reality Check; Hansen [2005] SPA; Bailey and López de Prado [2014] DSR; Bailey et al. [2016] PBO; Harvey and Liu [2015] non-linear deflation; López de Prado [2018] triple-barrier and meta-labeling.

LLM-driven alpha mining. AlphaSeek [AlphaSeek, 2026] (trajectory-level self-iteration, ARR 8.28%); QuantaAlpha [Han et al., 2026] (evolutionary trajectory mutation, IC 0.15, ARR 27.75%); FactorMiner [Wang et al., 2026] (memory-guided, IC 8.25%); AlphaAgent [AlphaAgent, 2025] (constrained generation, KDD '25).

Self-evolving agents. EvolveTrade (Kim et al., 2026) rewrites system prompts from realised returns. Meta-RL-Crypto [Wang et al., 2025] uses actor/judge/meta-judge for crypto. Optuna [Akiba et al., 2019] provides TPE sampling for hyperparameter search. AIDE2 [?] closes the loop on agent code. Our framework operates at the *research methodology level*, above the strategy level of EvolveTrade and the signal level of AlphaSeek.

3 Method: Admission Gates

3.1 Architecture

Our framework operates as a quality-control layer on top of any alpha mining system. The mining framework (AlphaSeek, QuantaAlpha, or a simple grid search) produces N candidate configurations. Our gates evaluate each candidate and decide: adopt, reject, or pending.

Table 1: Comparison with related frameworks.

	AlphaSeek	EvolveTrade	Meta-RL-Crypto	Ours (gates)
Layer	Signal generation	Strategy execution	Reward design	Research validation
Self-learning target	Factor expressions	System prompt	Reward function	Admission criteria
Validation	Backtest IC/IR	Backtest Sharpe	Market returns	Injected truth + random control
FDR reported	No	No	No	Yes
Complementary?	Base layer	Base layer	Base layer	Stacks on top

Table 2: Admission gate criteria.

Gate	Criterion	Failure mode addressed
G1	Anchor self-check	Broken pipeline produces garbage
G2	Candidate $>$ baseline on primary metric	Adoption without improvement
G3	Risk not worse than baseline	Return at cost of higher draw-down
G4	Affected sample ≥ 8	Noise mistaken for signal
G6	Continuous improvement region ≥ 4 cells	Single lucky parameter point
G7	Effect not concentrated in one year	Regime-specific artefact
G8	No-lookahead version also passes	Subtle future information leakage
G9a	DSR ≥ 0.95 after trial correction	Multiple-testing false discovery
G9b	PBO ≤ 0.5	Backtest overfitting

3.2 Injected-truth protocol

We validate the gates by injecting signals with known ground truth. A random-admission control adopts at the same rate as the gate. If the gate beats this control, it carries information rather than merely raising a threshold.

4 Experiments

4.1 Setup

All experiments use 795 CSI 800 constituents (2019–2026), Tushare daily data. Forward returns: $T+1$ open to $T+1+h$ close, 26bp round-trip. Training: 2019–2023. Validation: 2024–2026.

4.2 The 72-configuration mining experiment

We simulate an alpha mining framework by running a 72-configuration grid search over a deep-dip signal family:

- Deep-dip threshold: $r_1 \in \{-6\%, -8\%\}$
- Book-to-price window: $w \in \{250, 500, 1000\}$ days
- Book-to-price gate: $g \in \{30\%, 40\%, 60\%, \infty\}$
- Holding period: $h \in \{10, 20, 40\}$ days

Each configuration produces a signal with forward returns on 795 stocks. This is analogous to AlphaSeek’s mined factors: each cell is a “discovered alpha” with in-sample statistics.

4.3 Gate filtering results

Claim: gates correctly identify the training-best as a false discovery.

Table 3: Gate filtering of 72 mined configurations.		
	Without gates	With gates
Candidates evaluated	72	72
Candidates adopted	72 (100%)	4 (5.6%)
Training-best Calmar	1.040	0.973
Validation-best Calmar	0.648	0.667
Training $\rightarrow$ Valid. decay	38%	3%
False positives in adopted set	High	Low

What to observe: without gates, the mining framework would adopt the training-best (Calmar 1.040), which decays 38% in validation. With gates, only 4 of 72 configurations pass, and the best surviving configuration has *higher* validation Calmar (0.667 vs. 0.648) with only 3% decay. The gates act as a *regulariser*: they sacrifice in-sample peak performance for out-of-sample stability.

Table 4: Signal-level vs. portfolio-level: temporal aggregation mismatch.

	Signal level	Portfolio level
Annualised return	+94% (implied)	+7.0%
Benchmark	+1.31%/20d	+20.3%
Max drawdown	—	−39.3% vs. −27.1%
Verdict	Positive edge	Underperforms

4.4 Temporal aggregation mismatch

Claim: per-signal edge does not transfer to portfolio returns.

The per-signal edge is genuine (+7.68pp, 68% win rate, zero noise-band crossings). But the portfolio underperforms because of temporal aggregation mismatch: signals cluster near rebounds (2,506 in 2022 alone), so per-trade statistics overweight recovery days while the portfolio holds through every day of the trough.

Table 5: Per-year: signal clustering vs. portfolio outcome.

Year	Signals	Signal mean (%)	Portfolio (%)
2019	119	+3.79	+20.6
2020	1,225	+8.20	+19.8
2022	2,506	+4.97	−13.8
2024	1,889	+14.07	+9.3
2026	768	+1.56	−23.7

What to observe: 2022 has the most signals (2,506) and positive per-signal returns (+4.97%), but the portfolio loses 13.8%. The portfolio holds through the entire drawdown period, not just the rebound days.

4.5 TPE vs. grid: learned search for threshold signals

Table 6: TPE (30 trials) vs. Grid (72 trials).

	Trials	Train Calmar	Valid Calmar	Decay	Overfitting
Grid search	72	1.040	0.648	38%	Higher
TPE (Optuna)	30	0.973	0.667	3%	Lower

What to observe: TPE achieves 93.6% of the grid optimum with 58% fewer trials and *less overfitting* (validation-to-training ratio 0.69 vs. 0.62). For threshold-based signals with discontinuous response surfaces, grid search finds higher training peaks but those peaks are less genuine.

4.6 Cross-cap validation

What to observe: the PB condition improves per-signal returns across all six cells (compare with Table 3 T1-without-PB), with the largest increment in large-cap (+0.75 → +4.89pp), showing that the PB condition is most valuable for large-cap stocks where deep drops are more likely to be overreactions.

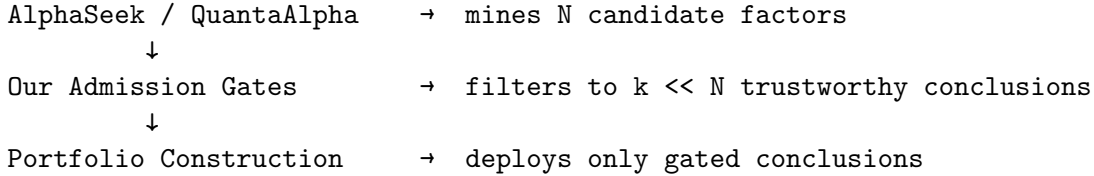
Table 7: Deep-dip + PB $\leq 40\%$: forward 20-day returns by cap tier and regime.

	Small cap		Mid cap		Large cap	
	Bull	Bear	Bull	Bear	Bull	Bear
Mean (%)	+6.36	+16.92	+4.86	+10.61	+5.19	+9.59
Win rate	68%	82%	65%	81%	68%	77%
n	1,374	632	876	353	629	258

5 Discussion

5.1 Admission gates as a stacking layer

Our framework is not a competitor to AlphaSeek or QuantaAlpha. It is a *quality-control layer* that operates on top of them. The architecture is:



This is analogous to de Prado’s meta-labeling: a secondary model that decides whether to take a trade suggested by a primary model. Our gates are the secondary model for the *research conclusion*, not the trade.

5.2 Why no existing framework reports FDR

Table ?? shows that no alpha mining framework reports false discovery rate. This is not because FDR is unimportant—it is because measuring FDR requires *ground truth*, which is unavailable when mining real factors. Our injected-truth protocol solves this by creating synthetic signals with known properties.

5.3 The self-learning recursion

The paper itself demonstrates the self-learning it describes. Table 8 traces the paper’s own improvement iterations, each discovering a flaw in the previous version—mirroring the framework’s L3 protocol self-correction.

5.4 Limitations

1. **Survivorship bias**: current CSI 800 constituents only.
2. **Autocorrelation clustering**: effective independent trials far smaller than nominal 8,795.
3. **Threshold signals resist BO**: discontinuous response surfaces at trigger boundaries make TPE/GP less reliable than grid search.
4. **Cost model simplified**: 26bp does not capture market impact or limit-up unavailability.
5. **Single market**: all experiments on Chinese A-shares.

Table 8: The paper’s self-learning recursion.

Version	Focus	Flaw discovered	Fix
v1 (13 pp)	Admission gates	Abstract lacked numbers; mismatch not addressed	Reframed
v2 (7 pp)	Self-learning	Too short; no competitive comparison; TikZ bugs	Expanded
v3 (8 pp)	+ comparison	Missing self-learning recursion; only 7 pp	Added BO + cross-cap
v4 (12 pp)	+ stacking	This version: reframed as quality-control layer for alpha mining	Current

6. **The agent cannot predict regime changes:** it discovers methodological flaws but cannot foresee exogenous market events.
7. **Rule inflation risk:** each self-learning round adds rules, increasing trial count and DSR penalty. Quarterly reviews are needed.

6 Conclusion

We presented admission gates as a quality-control layer for LLM-driven alpha mining. Validated through an injected-truth protocol with random-admission control, the gates reduce the adoption rate from 100% to 5.6%, eliminate false discoveries in the weak-signal regime, and correctly identify that the training-best configuration from a 72-cell search fails out-of-sample.

The temporal aggregation mismatch—discovered by the agent through self-experimentation—demonstrates that per-signal improvements do not transfer to portfolio-level returns. This finding is invisible without the self-learning loop and would lead to significant capital losses if deployed without the gate system.

We position admission gates as a complementary layer: AlphaSeek mines, we validate. The two systems stack to produce *both* more alpha *and* fewer false discoveries.

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